
AN0054- NB-IOT 常见网络问题

EigenCOMM Wireless Microcontroller

文档说明

本文档综合整理移芯 NB 芯片在使用过程中遇到的网络问题及相应解决办法。

目录

1. 开机搜网注册流程	3
2. EC616&EC616S 的搜网顺序	3
3. OOS 机制及恢复	4
4. RAI 快速释放 RRC 连接	6
5. 快速进入 PSM 模式的非标做法	7
6. EPCO 对 DNS 解析的影响	8
7. EC616&EC616S LOG 开关对灵敏度的影响	10
8. 网络被拒 RRC REJECT 的处理	10
9. 配置 UE 支持 R14 并获取 R14 能力	10
10. 版本	11
11. 关于我们	12

1. 开机搜网注册流程

EC616/EC616S 开机搜网注册流程图：

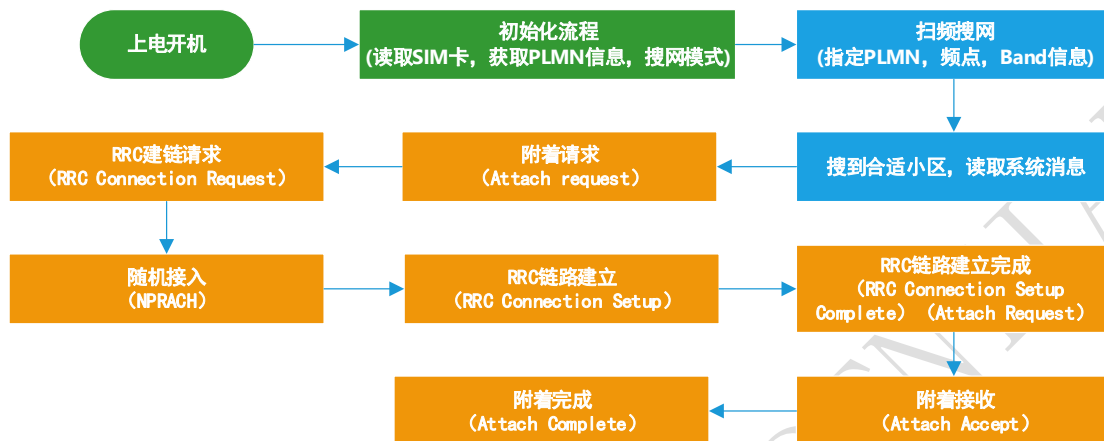


图 1-1 开机搜网注册流程

2. EC616&EC616S 的搜网顺序

当前 EC616&EC616S 的搜网顺序为：

按照优先级顺序，将Band列表下发给物理层，以加速驻网。Band优先级顺序如下：

- (1) 第一个优先频点（即最近一次驻留上的小区频点）所对应的 Band，需要提醒的是：该 Band 必须在用户设置的 Band 列表中，否则不予考虑；
 - a) 优先频点，关机保存；
 - b) 开机会检查当前SIM卡中的HPLMN和关机之前SIM卡的HPLMN是否一致，如果不一致，会清除优先频点信息；
- (2) SIM 卡所对应的优选 Band，客户可以在 dialPlmnCfgList 中自行添加、修改各个运营商的优选 Band 列表(psdial_plmn_cfg.c)，需要提醒的是：这些 Band 也必须在用户设置的 Band 列表中，否则不予考虑；

```

psdial_plmn_cfg.c (middleware\src)
psdial_plmn_cfg.c
Symbol Name (Alt+L)
include "psdial_plmn_cfg.h"
#define DEFINE_GLOBAL_VARIABLES_ //just for easy to find this position in SS
const DialPlmnCfgInfo dialPlmnCfgList[] =
{
  /*1 PLMN 2 eif 3 ipType 4 APN 5 pShortName 6 pLongName 7 pBand,*/
  /*8 pDefIp4Dns1, 9 pDefIp4Dns2, 10 pDefIp6Dns1, 11 pDefIp6Dns2*/
  /*{0x001, 0x001}, FALSE, DIAL_CFG_IPV4V6, NULL, NULL, NULL, NULL, {0,0,0,0,0,0,0,0}, //TEST SIM CARD*/
  /*CMCC*/
  /*{0x460, 0x000}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x002}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x004}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x006}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x008}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x010}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x012}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x014}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x016}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x018}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x020}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x022}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x024}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x026}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x028}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x030}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x032}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x034}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x036}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x038}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x040}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x042}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x044}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x046}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x048}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x050}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x052}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x054}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x056}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x058}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x060}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x062}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x064}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x066}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x068}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x070}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x072}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x074}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x076}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x078}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x080}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x082}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x084}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x086}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x088}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x090}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x092}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x094}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x096}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x098}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0A0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0A2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0A4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0A6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0A8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0AA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0AC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0AE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0B0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0B2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0B4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0B6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0B8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0BA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0BC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0BE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0C0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0C2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0C4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0C6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0C8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0CA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0CC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0CE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0D0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0D2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0D4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0D6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0D8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0DA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0DC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0DE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0E0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0E2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0E4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0E6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0E8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0EA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0EC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0EE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0F0}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0F2}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0F4}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0F6}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0F8}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0FA}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0FC}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*{0x460, 0x0FE}, FALSE, DIAL_CFG_IPV4V6, "cmmbiot", "CMCC", "CHINA MOBILE", {8,0,0,0,0,0,0,0},
  /*CTCC*/
  /*{0x460, 0x003}, TRUE, DIAL_CFG_IPV4V6, "ctnb", "CTCC", "CHINA TELECOM", {5,3,0,0,0,0,0,0},
  /*{0x460, 0x005}, TRUE, DIAL_CFG_IPV4V6, "ctnb", "CTCC", "CHINA TELECOM", {5,3,0,0,0,0,0,0},
  /*{0x460, 0x007}, TRUE, DIAL_CFG_IPV4V6, "ctnb", "CTCC", "CHINA TELECOM", {5,3,0,0,0,0,0,0},
  /*{0x460, 0x009}, TRUE, DIAL_CFG_IPV4V6, "ctnb", "CTCC", "CHINA TELECOM", {5,3,0,0,0,0,0,0},
  /*{0x460, 0x011}, TRUE, DIAL_CFG_IPV4V6, "ctnb", "CTCC", "CHINA TELECOM", {5,3,0,0,0,0,0,0},
}
  
```

- (3) 用户在 AT+ECBAND 中设置的 Band 顺序；

通过上述优化，UE可以加速驻留到对应的小区上。

(4) 加快搜网

建议根据运营商，部署使能部分频段；依据不同的中国运营商，可参照下表设置：

运营商	AT 命令	说明
中国移动	AT+ECBAND=0,8	仅使能 B8
中国电信	AT+ECBAND=0,5	仅使能 B5
中国联通	AT+ECBAND=0,3,8	仅使能 B3 和 B8

3. OOS 机制及恢复

OOS 机制可以用 AT+ECCFG=“PlmnSearchPowerLevel”,x 配置。截图如下下（参数 x 具体说明见 AT 手册。）：

“PlmnSearchPowerLevel” Set the PLMN search level when UE OOS;

Note:

a) Supported values: (0, 1, 2, 3, 4)

0 - OOS PLMN search interval: 30 sec, 1 min,

2 min

1 - OOS PLMN search interval: 5 min, 10 min, 15 min

2 - OOS PLMN search interval: 10 min, 30 min, 1 hour

3 - OOS PLMN search interval: 30 sec, then stop PLMN search, and let AT: AT+ECPLMNS to start PLMN search

4 – Don't perform PLMN search when OOS, let user to decide next action(whether perform PLMN search or not)

图 3-1 OOS 选项配置

目前的默认配置是 3，当发生 OOS 之后，会启动 30S 的定时器，定时器超时之后，会再进行一次搜网。假如此次搜网还是 OOS，那么就停止搜网，后续不再进行主动搜网。只有主动进行 CFUN0/CFUN1 或者重新 POWER OFF/ON/RESET/AT+ECPLMNS1 等才能进行重新搜网。

客户端应用在 UE 发生 OOS 之后，不需要进行主动的搜网，可以把 PlmnSearchPowerLevel 的值设置为 4，在 OOS 之后都不会触发主动搜网：AT+ECPCFG= “PlmnSearchPowerLevel”, 4

注意：发生了 OOS 之后，直接发送数据不会触发主动搜网。

Index	UETime	Message	PCTime
4354	01:1013:03:0344	Continue Frequency Scan in Band , Current EARFCN: 1406 , freqScore = 40	2022-03-24 22:55:36.307
4366	05:0051:06:0338	Continue Frequency Scan in Band , Current EARFCN: 1612 , freqScore = 40	2022-03-24 22:56:07.662
4404	08:0020:08:0323	Continue Frequency Scan in Band , Current EARFCN: 1818 , freqScore = 40	2022-03-24 22:56:38.059
4409	09:0928:07:0323	Frequency Scan in Band Finished!!!	2022-03-24 22:56:57.378
4411	09:0928:07:1053	CELL FOUND: cellDetected (0) , highLevelDone (0) , CELL (0 , 0) , RSRP (0) , RSRQ (0)	2022-03-24 22:56:57.378
4424	09:0928:09:0805	Setting error, failed to set psrnc	2022-03-24 22:56:57.378
4429	09:0929:01:1054	Start DeepSleep Timer, ticks = 30000, timerID = 0 - PLMN_PERIOD_OR_OOS_TIMER (0) , T3245 (1) , T3324 (2) , T3325 (3)	2022-03-24 22:56:57.378
4440	09:0929:02:0197	CCM REG , UE is OUT OF SERVICE	2022-03-24 22:56:57.378
4468	09:0929:03:1524	Hibernate failed reason = 4	2022-03-24 22:56:57.379
4501	09:0930:03:1760	PS DeepSleep Callback Excute	2022-03-24 22:56:57.410
4502	09:0930:03:1789	CCM , prepare to enter: 4 , HIB (4) / SLEEP2 (3) state	2022-03-24 22:56:57.410
4507	09:0930:04:0168	Uicc , prepare to enter: 4 , HIB (4) / SLEEP2 (3) state	2022-03-24 22:56:57.410
4508	09:0930:04:0180	PsIfTcpiEnterHibCallback state 4	2022-03-24 22:56:57.410
4520	09:0930:04:1087	Write Flash , Current Flag = 7 to Region = 2 1	2022-03-24 22:56:57.411
4530	09:0932:01:1173	Enter Hibernate 29915 ms PreSlp = 18 ms Wakeup SC = 322470154 minTimeL1 = 0 , CurrentSC = 321490197	2022-03-24 22:56:57.411
4532		Wakeup From Hibernate-0_0_0 , IntPending = 0x1	2022-03-24 22:57:27.354
4533		PS Tiny check , PS used: 2180 , reserved total: 2496	2022-03-24 22:57:27.354
4534	00:0000:00:0084	DCXO CBANK mismatch: 24 in table , 30 in FUSE	2022-03-24 22:57:27.354
4552	11:0321:02:0776	... Board: EC616S_EVK -- SDK Version: EC616S_SW_V001.037.p001.002 -- EVB Version: EC616S_HW_V1.0 -- Compile ...	2022-03-24 22:57:27.354
4553	11:0321:02:0837	AT port config: baudRate:9600 , dataBits: 8 , parity:0 (0 ->NONE , 1 ->ODD , 2 ->EVEN) , stopBits:1	2022-03-24 22:57:27.354
4562	11:0321:02:1488	PS Tiny check , PS used: 2180 , reserved total: 2496	2022-03-24 22:57:27.354
4568	11:0321:06:1767	DCXO CBANK mismatch: 24 in table , 30 in FUSE	2022-03-24 22:57:27.354
4640	11:0324:00:0256	PsIfRequestRecoverContextFromHib	2022-03-24 22:57:27.386
4644	11:0324:00:0569	AtSocketInit INIT	2022-03-24 22:57:27.386
4650	11:0324:00:1065	CAC DEV , CLOTPOWER wake from HIB	2022-03-24 22:57:27.386
4688	11:0324:04:1471	PSIF , suspend (1) / resume (0) : 1 LWIP	2022-03-24 22:57:27.386
4702	11:0324:06:1265	autoReg is disable	2022-03-24 22:57:27.387
4749	11:0325:03:1230	Net manager link up: CID 0 , bind to another CID 255	2022-03-24 22:57:27.387
4758	11:0325:04:1045	PsIfLinkUp netif enter oos	2022-03-24 22:57:27.387
4762	11:0325:06:1363	Wakeup From DeepSleep , Timer Restart , ticks = 20 , timerID = 0 - PLMN_PERIOD_OR_OOS_TIMER (0) , T3245 (1) , T3324 (...	2022-03-24 22:57:27.387
4785	11:0325:09:0684	Can't open NVM file: cisconfig.nvm	2022-03-24 22:57:27.387

图 3-2 OOS LOG

4. RAI 快速释放 RRC 连接

在终端产品应用中，很多应用场景对功耗敏感，需要想办法降低功耗，业务完成后需要快速的从 RRC 连接态返回到 IDLE 态，节省连接态的开销降低功耗。如果不做任何的干预，在现有的电信/移动 NB 网络条件下，从 RRC 到 IDLE 太可能会额外花费 15S~20S，极大的浪费了功耗。移芯 NB 平台提供多种方式来快速的释放 RRC 连接，进入 IDLE 态。举例如下：

- (1) 通过 AT+ECNBIOTRAI=1 AT 指令可以快速释放 RRC，减少 connect 状态时间，降低设备功耗。

30:0403:07:1678	DL Tpt STATUS INFO: MCS Index (414 / 100) , BLER (0 / 10000)	2022-04-02 21:11:14.066
30:0403:07:1689	UL Tpt STATUS INFO: MCS Index (1100 / 100) , BLER (0 / 10000)	2022-04-02 21:11:14.066
30:0423:00:0855	ATCMD_decode AT: AT+ECNBIOTRAI=1	2022-04-02 21:11:14.248
30:0423:01:1652	CESM , RAI flag: 1 , RAI_NO_INFO (0) / NO_UL_DL (1) / ONLY_DL (2) / REVD (3)	2022-04-02 21:11:14.248
30:0423:02:1587	Sending ESM_DATA_TRANSPORT:	2022-04-02 21:11:14.248
30:0423:03:0546	UllInformationTransfer-NB , len (15)	2022-04-02 21:11:14.248
30:0423:05:0103	AT CMD , RESP: OK	2022-04-02 21:11:14.248
30:0448:01:1762	Trigger PRACH , cause: 2 , CCCH (0) / NPDCCH_ORDER (1) / UL_DATA (2)	2022-04-02 21:11:14.474
30:0448:02:0244	Current CE = 0 , lastSuccCE = 0	2022-04-02 21:11:14.474
30:0448:02:1377	PRACH Transmitted , RA cause = 1 (contention free (0) / contention based (1)) , CarrierIndex = 0 (anchor carrier (0) / non-anchor carrier (1-15)) , SubCarrierIndex = 44 , R...	2022-04-02 21:11:14.474
30:0448:02:1393	CE level = 0 , RaContentionFlag = 1 (contention free (0) / contention based (1)) , CarrierIndex = 0 (anchor carrier (0) / non-anchor carrier (1-15)) , SubCarrierIndex = 44 , R...	2022-04-02 21:11:14.474
30:0451:02:0532	RAR_PDSCH received : NumOfRapid = 1	2022-04-02 21:11:14.534
30:0451:02:1862	RAR received: backOffIndicator = 0 , tmpCmti = 34816 , taCmd = 0 , rarGrant = 16386	2022-04-02 21:11:14.534
30:0453:02:0204	Msg3 Transmitted : crnti = 35178 , tmpCmti = 34816 , Msg3TransCnt = 1	2022-04-02 21:11:14.534
30:0454:00:0989	PRACH SUCC	2022-04-02 21:11:14.534
30:0458:07:1639	RrcConnectionRelease-NB:	2022-04-02 21:11:14.610
30:0458:09:0776	PSIF , suspend (1) / resume (0) : 1 LWIP	2022-04-02 21:11:14.610
30:0464:07:0639	MAC RESET , cause (3) : REEST (1) , REEST_RETRY (1) , EST_RETRY (2) , RETURN_IDLE (3) , SUSPEND (4)	2022-04-02 21:11:14.625
30:0465:01:1188	CELL (3686 , 243) , cellStatus (1) , SUITABLE (1) , BARRED (2) , PLMN_FORBIDDEN (3) , TA_FORBIDDEN (4) , selectedPmnIdx (1) , band (8)	2022-04-02 21:11:14.625
30:0465:01:1917	totalTxPower = 134 (mAs) , totalRxPower = 1613 (mAs) , totalTxTime = 1504 (ms) , totalRxTime = 107554 (ms)	2022-04-02 21:11:14.625
30:0465:03:0636	pagingCycle is 2560 (ms) , Edrx config: Edrx period is 40960 (ms) and Pw Length is 10240 (ms)	2022-04-02 21:11:14.625
30:0465:06:0523	Start T3324	2022-04-02 21:11:14.625
30:0465:08:1214	Checking timer T3324: 30000 , timerID = 0 - PLMN_PERIOD_OR_OOS_TIMER (0) , T3245 (1) , T3324 (2) , T3325 (3)	2022-04-02 21:11:14.625

图 4-1 快速释放 RRC

- (2) CTWING AEP 平台，发送数据可以携带 RAI 标记 - 参考 AT 命令手册

3.8.6 AT+CTM2MSEND 发送数据

发送业务数据

AT+CTM2MSEND	
设置命令 AT+CTM2MSEND=<data>[,<mode>]	响应 +CTM2MSEND: <msgID> OK 如果发生错误, 响应: +CTM2M ERROR: <err>
测试命令 AT+CTM2MSEND=?	响应 +CTM2MSEND: (支持列表 <data>),(支持列表 <mode>)

最大响应时间	OK
参数保存模式	40秒
Parameter	不保存
<data>	字符串类型 十六进制字符, 长度小于1024
<mode>	整型 0—CON 模式 1—NON 模式 2—NON 模式并且使能RAI 3—CON 模式并且使能RAI 缺省默认值为1

图 4-2

(3) CMCC OneNET 平台, 发送数据可以携带 RAI 标记 - 参考 AT 命令手册

3.7.15 AT+MIPLUPDATE 更新注册信息

该指令向 OneNET 发送更新注册信息

AT+MIPLUPDATE	
设置命令 AT+MIPLUPDATE=<ref>,<lifetime>,<withobjectflag>[,<raiflag>]	响应 OK 如果发生错误, 响应: +CIS ERROR: <err>
测试命令 AT+MIPLUPDATE=?	响应 +MIPLUPDATE: (支持列表 <ref>),(range of supported<lifetime>),(支持列表 <withobjectflag>),(支持列表<raiflag>) OK
最大响应时间	5秒
参数保存模式	不保存
参数	
<ref>	整型 +MIPLCREATE指令返回的客户端序号
<lifetime>	整型 保活时间, 如果为0, 表示使用默认的lifetime值
<withobjectflag>	整型 1 同时更新对象信息 0 不更新对象信息
<raiflag>	整型 0 (PS_SOCK_RAI_NO_INFO) 不开启RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) 开启RAI, 发送完立即释放 2 (PS_SOCK_ONLY_DL_FOLLOWED) 开启RAI, 收到回复后释放

图 4-3

5. 快速进入 PSM 模式的非标做法

(1) 前面提到可以通过 RAI 来快速释放 RRC 链接, 如果用户的场景需要快速的进入 PSM 模式, 减少 IDLE 模式下的时间和功耗的开销, 可以通过设定 T3324MaxValues 的值让 UE 在设定的时间内或者立即进入 PSM (设置为 0)。

用户可以通过 AT 指令: AT+ECCFG=" T3324MaxValueS", X 来配置 T3324 定时器的值。

当 X 为 0 时, RRC 释放后, 会立即进入 PSM。

1825767	26:0891:06:1595	UL Tpt STATIS INFO: MCS Index (1100 / 1000), BLER (0 / 10000)	2022-04-06 10:38:49.579
1825770	26:0908:07:1376	CONNECTED: SERV CELL (3686, 243) :RSRP (-86), RSRQ (-2), SNR (18)	2022-04-06 10:38:49.686
1825784	26:1011:00:1161	CONNECTED: SERV CELL (3686, 243) :RSRP (-87), RSRQ (-1), SNR (18)	2022-04-06 10:38:50.720
1825790	27:0067:06:1586	DL MEAS STATUS INFO: RSRP (-8621 / 100), RSRQ (-117 / 100), SNR (1885 / 100)	2022-04-06 10:38:51.501
1825791	27:0067:06:1597	DL Tpt STATIS INFO: MCS Index (425 / 100), BLER (0 / 10000)	2022-04-06 10:38:51.501
1825792	27:0067:06:1608	UL Tpt STATIS INFO: MCS Index (1100 / 1000), BLER (0 / 10000)	2022-04-06 10:38:51.501
1825794	27:0091:01:1866	CONNECTED: SERV CELL (3686, 243) :RSRP (-86), RSRQ (-1), SNR (18)	2022-04-06 10:38:51.730
1825801	27:0195:00:1146	CONNECTED: SERV CELL (3686, 243) :RSRP (-86), RSRQ (-1), SNR (19)	2022-04-06 10:38:52.667
1825815	27:0238:05:1492	RrcConnectionRelease-NB:	2022-04-06 10:38:53.206
1825825	27:0238:07:0616	PSIF, suspend (1) / resume (0) : 1 LWIP	2022-04-06 10:38:53.267
1825844	27:0244:04:1515	MAC RESET, cause (3), REEST (0), REEST_RETRY (1), EST_RETRY (2), RETURN_IDLE (3), SUSPEND (4)	2022-04-06 10:38:53.252
1825869	27:0244:09:0332	CELL (3686, 243), cellStatus (1), SUITABLE (1), BARRED (2), PLMN FORBIDDEN (3), TA FORBIDDEN (4), selectedPImid...	2022-04-06 10:38:53.252
1825872	27:0244:09:1020	totalTxPower = 111 (mAs), totalRPower = 1345 (mAs), totalTxTime = 1086 (ms), totalRxTime = 89723 (ms)	2022-04-06 10:38:53.252
1825876	27:0245:00:1616	pagingCyclic = 2560 (ms), Edrx config: Edrx period is 40960 (ms) and Ptw Length is 10240 (ms)	2022-04-06 10:38:53.252
1825894	27:0245:03:1581	T3324 assigned by NW is zero	2022-04-06 10:38:53.267
1825895	27:0245:03:1621	Start T3412	2022-04-06 10:38:53.267
1825899	27:0245:05:0620	Start DeepSleep Timer, ticks = 72000000, timerID = 8 - T3346 (5), T3402 (6), T3411 (7), T3412 (8)	2022-04-06 10:38:53.267
1825905	27:0245:06:0159	PSIF, suspend (1) / resume (0) : 0 LWIP	2022-04-06 10:38:53.267
1825928	27:0246:07:0783	CELL (3686, 243), MIB-NB, len (5)	2022-04-06 10:38:53.267
1825929	27:0246:07:0793	MIB-NB:	2022-04-06 10:38:53.267
1825958	27:0247:00:1459	Fail to store last camp cell info due to GPRS state (4), is ValidBitMap (true)	2022-04-06 10:38:53.268
1825985	27:0247:03:1652	Enter Psm Mode Suce!	2022-04-06 10:38:53.298
*****	*****	END!	

图 5-1 AT+ECCFG="T3324MaxValueS",0 配置

(2) T3324Max 的设定值解析可以参考 AT 命令手册，如下：

“T3324MaxValueS”

T3324的最大值控制开关（单位：秒）

说明：

- a) 支持的值：(0-16777215)
- b) 默认值16777215
- c) 如果T3324MaxValueS 小于 65535 并且网络配置的T3324的值等于或者大于T3324MaxValueS（或者网络侧没有配置T3324），T3324的值此时为T3324MaxValueS
- d) 如果T3324MaxValueS 小于 65535 并且网络配置的T3324的值小于T3324MaxValueS，T3324的值此时就是网络侧配置的值
- e) 如果T3324MaxValueS的值等于或者大于

65535，说明T3324MaxValueS不可用，直接采用网络侧配置的T3324的值

图 5-2 T3324Max 值的解析说明

(3) 通过设置 T3324MaxValues 来快速进入 PSM 是非标做法，主要是考虑到终端应用在不同的网络配置下需要快速进入 PSM 的场景。如果用户设定了 T3324MaxVualeS 值，则默认开启 PSM 模式，此时 CPSMS 的设定即使为 0 也会开启 PSM。

6. EPCO 对 DNS 解析的影响

(1) 通过 AT+ECCFG="Epc0", 0/1 可以配置是否使用 EPC0。配置为 0, 使用 EPC0, 在 attach ACCEPT 消息里会带有网络下发的 DNS 服务器。配置为 1, 不使用 EPC0, 则没有。

1817181	22:06:06:07...	UECapabilityInformation-NB:	2022-04-06 10:16:1...
1817185	22:06:06:09...	Receiving AUTHENTICATION_REQUEST:	2022-04-06 10:16:1...
1817259	22:06:25:07...	Sending AUTHENTICATION_RESPONSE:	2022-04-06 10:16:1...
1817262	22:06:25:07...	UllnformationTransfer-NB, len (19)	2022-04-06 10:16:1...
1817275	22:06:40:01...	Trigger PRACH, cause: 2, CCCH (0) / NPDCCH_ORDER (1) / UL_DATA (2)	2022-04-06 10:16:1...
1817277	22:06:40:01...	Current CE = 0, lastSuccCE = 0	2022-04-06 10:16:1...
1817278	22:06:40:01...	PRACH Transmitted, RA cause = UL_DATA	2022-04-06 10:16:1...
1817279	22:06:40:01...	CE level = 0, RaContentionFlag = 1 (contention free (0) / contention base...	2022-04-06 10:16:1...
1817280	22:06:43:02...	RAR_PDSCH received: NumOfRapid = 2	2022-04-06 10:16:1...
1817281	22:06:43:02...	RAR received: backOffIndicator = 0, tmpCrtm = 34154, taCmd = 0, rarGra...	2022-04-06 10:16:1...
1817289	22:06:45:02...	Msg3 Transmitted: crtmi = 34480, tmpCrtm = 34154, Msg3TransCnt = 1	2022-04-06 10:16:1...
1817298	22:06:46:01...	PRACH SUCC	2022-04-06 10:16:1...
1817319	22:06:51:06...	DllnformationTransfer-NB, len (16)	2022-04-06 10:16:1...
1817322	22:06:51:09...	Receiving SECURITY_MODE_COMMAND:	2022-04-06 10:16:1...
1817325	22:06:52:02...	Sending SECURITY_MODE_COMPLETE:	2022-04-06 10:16:1...
1817328	22:06:52:02...	UllnformationTransfer-NB, len (10)	2022-04-06 10:16:1...
1817362	22:06:62:02...	DllnformationTransfer-NB, len (12)	2022-04-06 10:16:1...
1817365	22:06:62:03...	Receiving ESM_INFORMATION_REQUEST:	2022-04-06 10:16:1...
1817379	22:06:62:08...	Sending ESM_INFORMATION_RESPONSE:	2022-04-06 10:16:1...
1817382	22:06:62:09...	UllnformationTransfer-NB, len (48)	2022-04-06 10:16:1...
1817415	22:06:90:00...	CONNECTED: SERV CELL (3686, 243): RSRP (-84), RSRQ (-6), SNR (19)	2022-04-06 10:16:1...
1817434	22:06:99:09...	DllnformationTransfer-NB, len (138)	2022-04-06 10:16:1...
1817440	22:07:00:00...	Receiving ATTACH_ACCEPT:	2022-04-06 10:16:1...
1817459	22:07:00:03...	PSIF, suspend (1) / resume (0): 0 LWIP	2022-04-06 10:16:1...
1817472	22:07:00:05...	Sending ATTACH_COMPLETE:	2022-04-06 10:16:1...
1817475	22:07:00:05...	UllnformationTransfer-NB, len (15)	2022-04-06 10:16:1...
1817512	22:07:00:09...	no all req	2022-04-06 10:16:1...
1817552	22:07:01:04...	Net manager link up: CID 0, bind to another CID 255	2022-04-06 10:16:1...
1817553	22:07:01:04...	Net manager, link up, CID: 0, IPV4: 100.118.183.230	2022-04-06 10:16:1...
1817554	22:07:01:04...	Net manager, link up, IPV4 PCO DNS: 211.136.17.107	2022-04-06 10:16:1...
1817555	22:07:01:04...	Net manager, link up, IPV4 PCO DNS: 211.136.20.203	2022-04-06 10:16:1...
1817567	22:07:01:04...	Netif status changed to: 3, NOT_DIAL (0) / DEACTIVATED (1) / OOS (2) / ...	2022-04-06 10:16:1...
1817632	22:07:06:08...	DllnformationTransfer-NB, len (24)	2022-04-06 10:16:1...
1817635	22:07:06:09...	Receiving EMM_INFORMATION:	2022-04-06 10:16:1...

图 6-1 EPCO=0 的 attach accept 消息

1820804	31:0128:0...	Trigger PRACH, cause: 2, CCCH (0) / NPDCCH_ORDER (1) / UL_DA...	2022-04-06 10:17...
1820806	31:0128:0...	Current CE = 0, lastSuccCE = 0	2022-04-06 10:17...
1820807	31:0128:0...	PRACH Transmitted, RA cause = UL_DATA	2022-04-06 10:17...
1820808	31:0128:0...	CE level = 0, RaContentionFlag = 1 (contention free (0) / contenti...	2022-04-06 10:17...
1820809	31:0131:0...	RAR_PDSCH received: NumOfRapid = 2	2022-04-06 10:17...
1820810	31:0131:0...	RAR received: backOffIndicator = 0, tmpCrtm = 35151, taCmd = 0 ...	2022-04-06 10:17...
1820818	31:0133:0...	Msg3 Transmitted: crtmi = 34141, tmpCrtm = 35151, Msg3TransCn...	2022-04-06 10:17...
1820827	31:0134:0...	PRACH SUCC	2022-04-06 10:17...
1820848	31:0139:0...	DllnformationTransfer-NB, len (16)	2022-04-06 10:17...
1820851	31:0139:0...	Receiving SECURITY_MODE_COMMAND:	2022-04-06 10:17...
1820854	31:0140:0...	Sending SECURITY_MODE_COMPLETE:	2022-04-06 10:17...
1820857	31:0140:0...	UllnformationTransfer-NB, len (10)	2022-04-06 10:17...
1820891	31:0150:0...	DllnformationTransfer-NB, len (12)	2022-04-06 10:17...
1820894	31:0150:0...	Receiving ESM_INFORMATION_REQUEST:	2022-04-06 10:17...
1820900	31:0151:0...	Sending ESM_INFORMATION_RESPONSE:	2022-04-06 10:17...
1820912	31:0151:0...	UllnformationTransfer-NB, len (48)	2022-04-06 10:17...
1820945	31:0178:0...	CONNECTED: SERV CELL (3686, 243): RSRP (-82), RSRQ (-6), SN...	2022-04-06 10:17...
1820965	31:0201:0...	DllnformationTransfer-NB, len (95)	2022-04-06 10:17...
1820970	31:0201:0...	Receiving ATTACH_ACCEPT:	2022-04-06 10:17...
1820989	31:0202:0...	PSIF, suspend (1) / resume (0): 0 LWIP	2022-04-06 10:17...
1821002	31:0202:0...	Sending ATTACH_COMPLETE:	2022-04-06 10:17...
1821005	31:0202:0...	UllnformationTransfer-NB, len (15)	2022-04-06 10:17...
1821042	31:0202:0...	no all req	2022-04-06 10:17...
1821083	31:0203:0...	Net manager link up: CID 0, bind to another CID 255	2022-04-06 10:17...
1821084	31:0203:0...	Net manager, link up, CID: 0, IPV4: 100.84.212.174	2022-04-06 10:17...
1821103	31:0203:0...	DL MEAS STATUS INFO: RSRP (-8200 / 100), RSRQ (-700 / 100), SN...	2022-04-06 10:17...
1821104	31:0203:0...	DL Tpt STATUS INFO: MCS Index (407 / 100), BLER (0 / 10000)	2022-04-06 10:17...
1821105	31:0203:0...	UL Tpt STATUS INFO: MCS Index (1100 / 100), BLER (0 / 10000)	2022-04-06 10:17...
1821129	31:0204:0...	Netif status changed to: 3, NOT_DIAL (0) / DEACTIVATED (1) / OO...	2022-04-06 10:17...
1821168	31:0208:0...	DllnformationTransfer-NB, len (24)	2022-04-06 10:17...
1821171	31:0208:0...	Receiving EMM_INFORMATION:	2022-04-06 10:17...
1821208	31:0278:0...	CONNECTED: SERV CELL (3686, 243): RSRP (-82), RSRQ (-1), SN...	2022-04-06 10:17...
1821215	31:0380:0...	CONNECTED: SERV CELL (3686, 243): RSRP (-82), RSRQ (-1), SN...	2022-04-06 10:17...
1821221	31:0403:0...	DL MEAS STATUS INFO: RSRP (-8200 / 100), RSRQ (-116 / 100), SN...	2022-04-06 10:17...

图 6-2 EPCO=1 的 attach accept 消息

(2) 配置的 DNS 服务器使用优先级如下，假如网络没有配置 DNS Server，用户也没有使用 AT 命令配置 DNS Server，那么 UE 最后只会使用默认的 DNS Server。

- DNS，网络配置的，在 PCO 内 (ATTACH ACCEPT)
- DNS，用户配置通过 AT+ECDNSCFG 指令 // 重启保存
- DNS，固定配置在文件 psdail_plmn_cfg.c 中，根据 SIM 卡的 IMSI

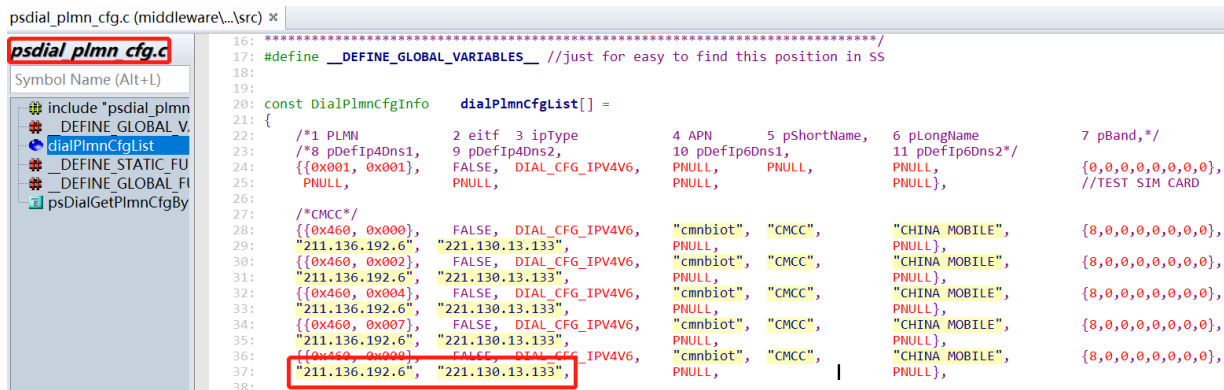


图 6-3 默认 DNS 配置

- (4) 在入库测试以及认证测试过程中，EPCO 的配置需要通过 AT 命令修改为“1”

7. EC616&EC616S LOG 开关对灵敏度的影响

EC616/EC616S/EC616L 等产品 SDK 从 V001.038+P02 版本以后默认 UART0 LOG 是关闭状态。如果客户 PCB 设计不好，UART0 的 3M log 打印可能会影响接收灵敏度。

- (1) AT 命令打开 LOG: AT+ECPCFG="logCtrl",1; 打开 LOG: AT+ECPCFG="logCtrl",0
(2) V001.038+P02 之前的版本默认是打开的，用户可以自己选择是否可以关闭。

8. 网络被拒 RRC REJECT 的处理

UE 注册网络被拒可能存在多种情况，除了网络本身的情况，例如网络拥塞导致的接入困难之外，优先需要排查 SIM 卡，SIM 卡主要从如下几个方面排查：

- (1) 卡没有开通业务（或者是回收的卡号等）
- (2) 如果已经使用过，咨询运营商是否已经欠费
- (3) 机卡绑定，SIM 卡已经在其它 UE 上使用过
- (4) 软件上是否做过不兼容的升级/降级，造成 NV 不兼容导致网络信息解析错误
- (5) 不是 NB 卡，有可能使用的是 4G 卡
- (6) 如果是同一批次的新卡，从未使用，建议直接采用对比机验证

9. 配置 UE 支持 R14 并获取 R14 能力

- (1) 目前 EC616/EC616S 默认支持 R13，可以使用 AT 指令开启支持 R14：

AT+CFUN=0

AT+ECCFG="RelVersion",14

- (2) 查询 UE R14 的能力采用 AT 命令：

AT+ECNBR14

+ECNBR14: UeRelVersion,14,UeR14UpRai,0

+ECNBR14: TwoHarq,0,R14UpRai,0,NonAnchorNPRACH,0,NonAnchorPaging,0,CpReest,0

10. 版本

版本	日期	备注
Draft	2022-04-02	

11. 关于我们

上海移芯通信科技有限公司坐落于上海张江，公司于 2017 年 2 月成立，从事蜂窝移动通信芯片的研发和销售。公司产品主要应用于广域物联网通信领域，致力于设计最具性价比的蜂窝物联网芯片。公司团队在蜂窝通信芯片上有着辉煌历史和丰富经验。公司所有核心技术全部自研，包括算法&架构、射频、基带、SoC、协议栈软件、平台&应用软件和硬件方案等。

移芯通信首款自主研发的超低功耗 NB-IoT 芯片 EC616 于 2019 年中量产，被绝大部分头部模组企业采用，现已大批量应用于全国各区域各行业。下一代超高集成度 NB-IoT 芯片 EC616S 于 20 年底量产，超低功耗 4G Cat.1 芯片 EC618 工程样片阶段，5G 芯片正在研发中。

上海移芯通信科技有限公司

地址：中国上海市浦东新区祥科路 298 号佑越国际 1 幢 7 层

邮编：201210

电话：

电邮：

网址：<http://www.eigencomm.com>

技术支持窗口

电邮：support@eigencomm.com